Cutaneous Drug Reactions

Greg Gayer, PhD

Chapter 60: Harrison's Principles of Internal Medicine

Introduction

- Cutaneous drug reactions
  - 2.2-10/1000 hospitalized patients.
  - Reactions time few days to 4 weeks after initiation of therapy.
  - Most common:
    - Rash
      - Morbilliform rash (91%)
      - Urticaria (6%)
    - Common Drugs
      - Antimicrobials
      - Radiocontrast
      - NSAIDS
  - Most Serious severe hypersensitivity
    - 1/1000 to 2 Mill.
  - Other – pruritus, pigmentation, nail or hair disorders, psoriasis, bullous pemphigoid, photosensitivity, and cutaneous neoplasms.

- Most at risk
  - Elderly
  - Patients with autoimmune disease
  - Hematopoietic stem cell transplant recipients,
  - EBV or HIV exposure

In the United States, more than 4 billion prescriptions for >60,000 drug products are dispensed annually. Hospital inpatients alone annually receive about 120 million courses of drug therapy, and half of adult Americans receive prescription drugs on a regular outpatient basis. Adverse effects of a prescription medication may result in 4.5 million urgent or emergency care visits and over 7000 deaths each year in the United States. Many patients use over-the-counter medicines that may cause adverse cutaneous reactions.

Several recent prospective studies reported that acute cutaneous reactions to drugs affect between 2.2 and 10 per 1000 hospitalized patients. Reactions usually occur a few days to 4 weeks after initiation of therapy.

In a series of 48,005 inpatients over a 20-year period, morbilliform rash (91%) and urticaria (6%) were the most frequent skin reactions, and antimicrobials, radiocontrast, and nonsteroidal anti-inflammatory drugs (NSAIDs) were the most common drug associations. Severe hypersensitivity reactions to medications have been reported to occur in between 1 in 1000 to 2 per million users, depending on the reaction type. Although rare, severe cutaneous reactions to drugs have an important impact on health because of significant sequelae; in addition, they may require hospitalization, increase the duration of hospital stay, or be life threatening. Some populations are at increased risk of drug reactions, including elderly patients, patients with autoimmune disease, hematopoietic stem cell transplant recipients, and those with acute Epstein-Barr virus (EBV) or human immunodeficiency virus (HIV) infection. The pathophysiology underlying this association is unknown but may be related to immune dysregulation. Individuals with advanced HIV disease (e.g., CD4 T lymphocyte count <200 cells/μL) have a 40- to 50-fold increased risk of adverse reactions to sulfamethoxazole **(Chap. 202)** and increased risk of severe hypersensitivity reactions.

In addition to acute eruptions, a variety of skin diseases can be induced or exacerbated by prolonged use of drugs (e.g., pruritus, pigmentation, nail or hair disorders, psoriasis, bullous pemphigoid, photosensitivity, and even cutaneous neoplasms). These drug reactions are not frequent; however, neither their incidence nor their impact on public health has been evaluated.

Pathogenesis

- Non immunologic: pigment changes, hair loss, lipodystrophy
- Immunologic
  - Immediate
  - Immune complex dependent
  - Delayed hypersensitivity

TABLE 60-1

Classification of Adverse Drug Reactions Based on Immune Pathway

TYPE	KEY PATHWAY	KEY IMMUNE MEDIATORS	ADVERSE DRUG REACTION TYPE
Type I	IgE	IgE	Urticaria, angioedema, anaphylaxis
Type II	IgG-mediated cytotoxicity	IgG	Drug-induced hemolysis, thrombocytopenia (e.g., penicillin)
Type III	Immune complex	IgG + antigen	Vasculitis, serum sickness, drug-induced lupus
Type IVa	T lymphocyte–mediated macrophage inflammation	IFN-γ, TNF-α T_H1 cells	Tuberculin skin test, contact dermatitis
Type IVb	T lymphocyte–mediated eosinophil inflammation	IL-4, IL-5, IL-13 T_H2 cells Eosinophils	DIHS Morbilliform eruption
Type IVc	T lymphocyte–mediated cytotoxic T lymphocyte inflammation	Cytotoxic T lymphocytes Granzyme Perforin Granulysin (SJS/TEN] only)	SJS/TEN Morbilliform eruption
Type IVd	T lymphocyte–mediated neutrophil inflammation	CXCL8, IL-17, GM-CSF Neutrophils	AGEP

Abbreviations: AGEP, acute generalized exanthematous pustulosis; DIHS, drug-induced hypersensitivity syndrome; GM-CSF, granulocyte-macrophage colony-stimulating factor; IFN, interferon; IL, interleukin; SJS, Stevens-Johnson syndrome; TEN, toxic epidermal necrolysis; TNF, tumor necrosis factor.

NONIMMUNOLOGIC DRUG REACTIONS

Examples of nonimmunologic drug reactions are pigmentary changes due to dermal accumulation of medications or their metabolites, alteration of hair follicles by antimetabolites and signaling inhibitors, and lipodystrophy associated with metabolic effects of antiHIV medications. These side effects are predictable and sometimes can be prevented.

Immunologic

Immediate Reactions

Immediate reactions depend on the release of mediators of inflammation by tissue mast cells or circulating basophils. These mediators include histamine, leukotrienes, prostaglandins, bradykinins, platelet activating factor, enzymes, and proteoglycans. Drugs can trigger mediator release either directly (“anaphylactoid” reaction) or through IgE specific antibodies. These reactions usually manifest in the skin and gastrointestinal, respiratory, and cardiovascular systems (Chap. 353). Primary symptoms and signs include pruritus, urticaria, nausea, vomiting, abdominal cramps, bronchospasm,

laryngeal edema, and, occasionally, anaphylactic shock with hypotension and death. They occur within minutes of drug exposure. NSAIDs, including aspirin, and radiocontrast media are frequent causes of direct mast cell degranulation or anaphylactoid reactions, which can occur on first exposure. Penicillins and muscle relaxants used in general anesthesia are the most frequent causes of IgE dependent reactions to drugs, which require prior sensitization. Release of mediators is triggered when polyvalent drug protein conjugates crosslink IgE molecules fixed to sensitized cells. Certain routes of administration favor different clinical patterns (e.g., gastrointestinal effects from oral route, circulatory effects from intravenous route).

Immune Complex Dependent

Serum sickness is produced by tissue deposition of circulating immune complexes with consumption of complement. It is characterized by fever, arthritis, nephritis, neuritis, edema, and an urticarial, papular, or purpuric rash (Chap. 363). First described following administration of nonhuman sera, it currently occurs in the setting of monoclonal antibodies and similar medications. In classic serum sickness, symptoms develop 6 or more days after drug exposure, the latent period representing the time needed to synthesize antibody. Vasculitis, a relatively rare complication of drugs, may also be a result of immune complex deposition (Chap. 363). Penicillin, cefaclor, amoxicillin, trimethoprim/sulfamethoxazole, and monoclonal antibodies such as infliximab, rituximab, and omalizumab may be associated with clinically similar "serum sickness–like" reactions (SSLR). The mechanism of this reaction is unknown but is unrelated to immune complex formation and complement activation, and

systemic involvement is rare. Whereas serum sickness most commonly occurs in adults, SSLR is more frequently observed in children.

Delayed Hypersensitivity

While not completely understood, delayed hypersensitivity directed by drug specific T cells is an important mechanism underlying the most common drug eruptions, that is, morbilliform eruptions, and also rare and severe forms such as drug induced hypersensitivity syndrome (DIHS) (also known as drug rash with eosinophilia and systemic symptoms [DRESS]), acute generalized exanthematous pustulosis (AGEP), StevensJohnson syndrome (SJS), and toxic epidermal necrolysis (TEN) (Table 601). Drug specific T cells have been detected in these types of drug eruptions. In TEN, skin lesions contain T lymphocytes reactive to autologous lymphocytes and keratinocytes in a drug specific, human leukocyte antigen (HLA)restricted, and perforin/granzy memediated pathway. In the case of carbamazepine, studies have identified cytotoxic T lymphocytes (CTLs) reactive to carbamazepine that use highly restricted Valpha and Vbeta TCR repertoires in patients with carbamazepine hypersensitivity that are not found in carbamazepine tolerant individuals. The mechanism(s) by which medications result in Tcell activation is unknown. Two hypotheses prevail: first, that the antigens driving these reactions may be the native drug itself or components of the drug covalently complexed with endogenous proteins, presented in association with HLA molecules to T cells through the classic antigen presentation pathway or, alternatively, through direct interaction of the drug/metabolite with the TCR or peptide loaded HLA (e.g., the pharmacologic interaction of drugs with immune receptors, or pi hypothesis). Recent xray crystallography data characterizing binding between specific HLA molecules to drugs known to cause hypersensitivity reactions demonstrate unique alterations to the MHC peptide binding groove, suggesting a molecular basis for Tcell activation in the development of hypersensitivity reactions.

Pustular Eruptions

AGEP is a rare reaction pattern affecting 3–5 people per million per year. It is thought to be secondary to medication exposure in >90% of cases (Fig. 6012). Patients typically present with diffuse erythema or erythroderma, as well as high spiking fevers and leukocytosis with neutrophilia. One to two days later, innumerable pinpoint pustules develop overlying the erythema. The pustules are most pronounced in body fold areas; however, they may become generalized and, when coalescent, can lead to superficial erosion. In such

cases, differentiating the eruption from SJS in its initial stages may be difficult, although in AGEP, any erosions tend to be more superficial, and prominent mucosal involvement is lacking. Skin biopsy shows collections of neutrophils and sparse necrotic keratinocytes in the upper part of the epidermis, unlike the fullthickness epidermal necrosis that characterizes SJS. Before the pustules appear, AGEP may also mimic DIHS due to the prominent fever and erythroderma.
Acute generalized exanthematous pustulosis.

Immediate Reactions

- Immediate reactions (minutes) depend on the release of mediators of inflammation by tissue mast cells or circulating basophils.
  - histamine, leukotrienes, prostaglandins, bradykinins, platelet activating factor, enzymes, and proteoglycans.
- Drugs can trigger mediator release either directly ("anaphylactoid" reaction) or through IgE specific antibodies. These reactions usually manifest in the skin and gastrointestinal, respiratory, and cardiovascular systems.
- Primary symptoms and signs include pruritus, urticaria, nausea, vomiting, abdominal cramps, bronchospasm, laryngeal edema, and, occasionally, anaphylactic shock with hypotension and death.
- Common drugs: NSAIDs, including aspirin, and radiocontrast media. Penicillin's and muscle relaxants used in general anesthesia (succinylcholine, vecuronium, pancuronium and atracurium) are the most frequent causes of IgE dependent reactions to drugs, which require prior sensitization. Release of mediators is triggered when polyvalent drug protein conjugates crosslink IgE molecules fixed to sensitized cells.

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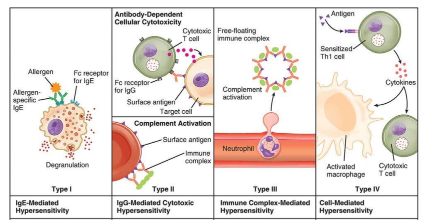

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Type I (Immediate Hypersensitivity)

Type I hypersensitivity is an allergic reaction provoked by re-exposure to a specific antigen called an allergen. IgE antibodies bind to receptors (FcεRI) found on the surface of tissue mast cells and blood basophils and sensitize them 2. Upon secondary exposure, the IgE on the surface of the sensitized mast and basophil cells becomes cross-linked. When two IgE receptors are cross-linked, signal transduces through the γ chains of the receptor leading to an influx of calcium, which initiates both degranulation and release of mediators such as histamines, leukotriene, and prostaglandin into the extracellular environment within minutes, producing allergic reactions. As the reaction occurs within a few minutes, type I hypersensitivity is also known as immediate hypersensitivity. Some examples are anaphylaxis, food allergies, drug allergies and dust allergies.

Immune Complex Dependent

- "Serum sickness" is produced by tissue deposition of circulating immune complexes with consumption of complement.
- Symptoms: fever, arthritis, nephritis, neuritis, edema, and an urticarial, papular, or purpuric rash (see Harrison's Chap. 363) that develop 6 or more days after drug exposure. Vasculitis, a relatively rare complication of drugs may occur. Whereas serum sickness most commonly occurs in adults, SSLR is more frequently observed in children.
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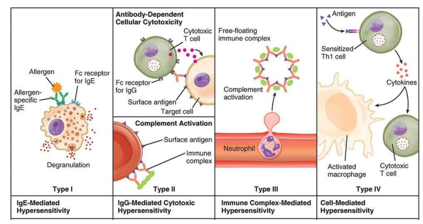

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Delayed Hypersensitivity

- Very serious types of reactions directed by drug specific T cells
- Symptoms range from morbilliform eruptions to severe drug induced hypersensitivity syndrome (DIHS) (also known as drug rash with eosinophilia and systemic symptoms [DRESS]), acute generalized exanthematous pustulosis (AGEP), Stevens-Johnson syndrome (SJS), and toxic epidermal necrolysis (TEN). The later will be covered later
    - Pathology involves drug specific T cells. In TEN, skin lesions contain T lymphocytes reactive to autologous lymphocytes and keratinocytes in a drug specific, human leukocyte antigen (HLA)restricted, and perforin/granzy mediated pathway. The mechanism(s) by which medications result in Tcell activation is unknown.
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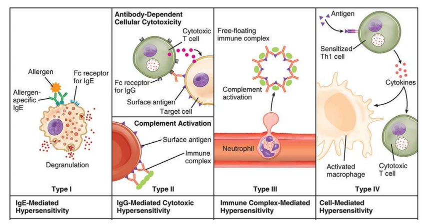

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Genetics Factors and Cutaneous Drug Rx.

- Pharmacokinetic
  - Mutations in metabolic enzymes
  - Excessive serum levels of drugs
- HLA haplotypes and hypersensitivity
  - Eg. Abacavir hypersensitivity – HLAB*57:01
  - Eg. Han Chinese patients
    - Carbamazepine –SJS/TNS –HLAB* 15:02
    - Allopurinol -- SJS, TEN, or DIHS HLAB* 58:01
- Genetic Screening before medications

Genetic determinants may predispose individuals to severe drug reactions by affecting either drug metabolism or immune responses to drugs. Polymorphisms in cytochrome P450 enzymes, drug acetylation, methylation (such as thiopurine methyltransferase activity and azathioprine), and other forms of metabolism (such as glucose6phosphate dehydrogenase and dapsone) may increase susceptibility to drug toxicity or underdosing and increase risk for medication interactions, highlighting a role for differential pharmacokinetic or pharmacodynamic effects. The value of routine screening of P450 enzymes for prediction of cutaneous reactions has not been determined, though its cost effectiveness in certain populations (e.g., patients with seizure disorder, depression) as well as patients considering specific therapies (e.g., tamoxifen, warfarin) has been suggested.

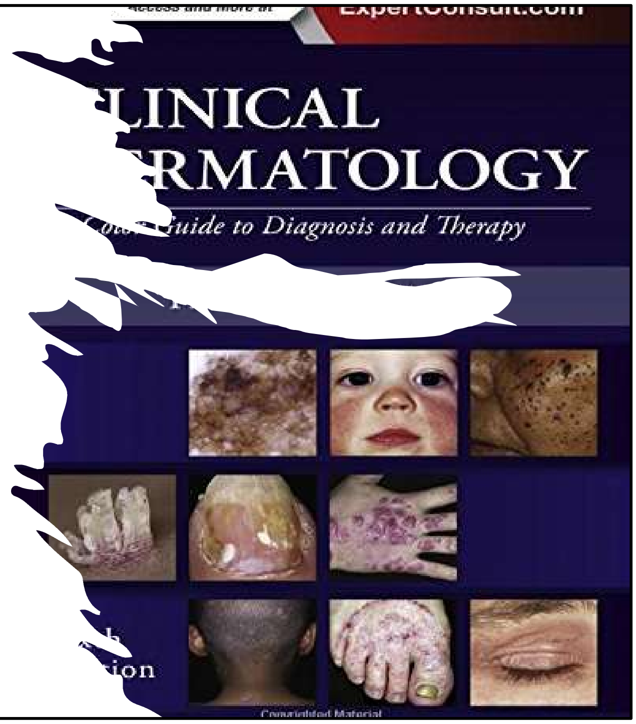
Associations between drug hypersensitivities and HLA haplotypes suggest a key role for immune mechanisms, especially those leading to skin involvement. Hypersensitivity to the antiHIV medication abacavir is strongly associated with HLAB* 57:01 (Chap. 202). In Taiwan, within a homogeneous Han Chinese population, a strong association was observed between SJS/TEN (but not DIHS) related to carbamazepine and HLAB* 15:02. In the same population, a strong association was found between HLAB* 58:01 and SJS, TEN, or DIHS related to allopurinol. These associations are drug and phenotype specific; that is, HLA specific Tcell stimulation by medications leads to distinct reactions. However, while this genetic association is strong, it is not sufficient to cause severe drug hypersensitivity reactions.

GLOBAL CONSIDERATIONS

Recognition of HLA associations with drug hypersensitivity has resulted in recommendations to screen high risk populations. Genetic screening for HLAB*57:01 to prevent abacavir hypersensitivity, which carries a 100% negative predictive value when patch test confirmed and 55% positive predictive value generalizable across races, is becoming the clinical standard of care worldwide (number needed to treat = 13). The U.S. Food and Drug Administration has recommended HLAB* 15:02 screening of Asian individuals prior to a new prescription of carbamazepine. The American College of Rheumatology has recommended HLAB* 58:01 screening of Han Chinese patients prescribed allopurinol. To date, screening for a single HLA (but not multiple HLA haplotypes) in specific populations has been determined to be cost effective (e.g., HLAB* 1301 screening in Chinese patients with leprosy treated with dapsone). Genetic testing for specific HLA haplotypes and functional screening for TCR repertoire to identify patients at risk is becoming more widely available and heralds the era of personalized medicine and pharmacogenomics.

Clinical Presentations of Cutaneous Drug Rx

- Exacerbating Cutaneous Reactions
- Photosensitivity Eruptions
- Pigmentation Changes
- Drug-induced Hair Disorders
- Drug-induced Nail Disorders
- Warfarin Skin Necrosis
- Toxic erythema of chemotherapy and other chemotherapy rx.
- Immune Cutaneous Rx
  - Common
  - Severe



Cutaneous Reactions: Exacerbation or Induction of Dermatologic Diseases

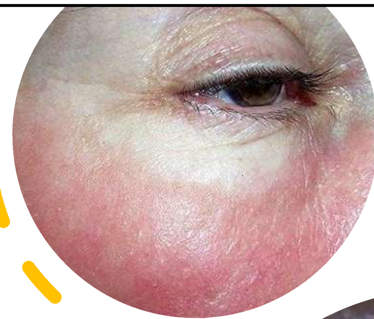
- Plaque psoriasis worsens sometimes with
  - NSAIDs, lithium, beta blockers, tumor necrosis factor (TNF) antagonists, interferon (IFN) α, and ACE inhibitors
- Pustular psoriasis worsens sometimes worsens with
  - Antimalarials or glucocorticoid withdrawal
- Acne
  - glucocorticoids, androgens, lithium, and antidepressants
- Acne like eruptions
  - epidermal growth factor receptor (EGFR) antagonists, mitogen activated protein kinase (MEK) inhibitors
- Subacute cutaneous lupus erythematosus
  - thiazide diuretics, proton pump inhibitors, TNF inhibitors, terbinafine, and minocycline
- Dermatomytositis
  - TNF inhibitors or capecitabine; hydroxyurea

NONIMMUNE CUTANEOUS REACTIONS

Exacerbation or Induction of Dermatologic Diseases

A variety of drugs can exacerbate preexisting diseases or induce—or unmask—a disease that may or may not disappear after withdrawal of the inducing medication. For example, NSAIDs, lithium, beta blockers, tumor necrosis factor (TNF) antagonists, interferon (IFN) α, and angiotensinconverting enzyme (ACE) inhibitors can exacerbate plaque psoriasis, whereas antimalarials and withdrawal of systemic glucocorticoids can worsen pustular psoriasis. The situation of TNFα inhibitors is unusual, as this class of medications is used to treat psoriasis; however, they may induce psoriasis (especially palmoplantar) in patients being treated for other conditions. Acne may be induced by glucocorticoids, androgens, lithium, and antidepressants. Follicular papular or pustular eruptions of the face and trunk resembling acne frequently occur with epidermal growth factor receptor (EGFR) antagonists, mitogen activated protein kinase (MEK) inhibitors, and other targeted inhibitors. With EGFR antagonists, the severity of the eruption correlates with a better anticancer effect. This rash is typically responsive to and prevented by tetracycline antibiotics. Several medications induce or exacerbate autoimmune disease. Checkpoint inhibitors induce a wide array of systemic autoimmune reactions, including in skin. Interleukin (IL) 2, IFNα, and antiTNFα are associated with new onset systemic lupus erythematosus (SLE). Drug induced lupus is classically marked by antinuclear and antihistone antibodies and, in some cases, antidoublestranded DNA (Dpenicillamine, antiTNFα) or perinuclear antineutrophil cytoplasmic antibodies (pANCA) (minocycline). Subacute cutaneous lupus erythematosus (SCLE) can be induced by a growing list of drugs, including thiazide diuretics, proton pump inhibitors, TNF inhibitors, terbinafine, and minocycline. Drug induced dermatomyositis may rarely occur with TNF inhibitors or capecitabine; hydroxyurea can induce skin findings of dermatomyositis. IFN and TNF inhibitors, as well as checkpoint inhibitors, can induce granulomatous disease and sarcoidosis. Autoimmune blistering diseases may be drug induced as well: pemphigus by Dpenicillamine and ACE inhibitors; bullous pemphigoid by DPP4 inhibitors, furosemide, and PD1 inhibitors; and linear IgA bullous dermatosis by vancomycin. Other medications may cause highly specific cutaneous reactions. Gadolinium contrast has been associated with nephrogenic systemic fibrosis, a condition of sclerosing skin with rare internal organ involvement; advanced renal compromise may be an important risk factor. Granulocyte colonystimulating factor, azacitidine, alltransretinoic acid, the FLT3 inhibitor class of drugs, and rarely levamisole contaminated cocaine may induce neutrophilic dermatoses. In this setting, the hypothesis that a drug may be responsible should always be considered, even after the treatment is complete. In addition, reactions may develop in cases of longterm medication therapy due to changes in dosing or host metabolism. Resolution of the cutaneous reaction may be delayed upon discontinuation of the medication.

Photosensitivity Reactions



- Drugs combine with UV radiation
  - Drug photoexcitation/photoconversion - multidirectional biological reactions.
    - Increased oxidative stress, inflammation, and changes in melanin synthesis.
    - Symptoms include: erythema, swelling, blisters, exudation, peeling, burning, itching, and hyperpigmentation of the skin.
  - Common drugs include:
    - fluoroquinolones, tetracycline antibiotics, and trimethoprim/sulfamethoxazole. Other drugs less frequently implicated are chlorpromazine, thiazides, NSAIDs, and BRAF inhibitors, voriconazole

PMCID: PMC8401619 DOI: 10.3390/ph14080723

Photosensitivity eruptions are usually most marked in sun exposed areas, but they may extend to sun protected areas. The mechanism is almost always phototoxicity. Phototoxic reactions resemble sunburn and can occur with first exposure to a drug. Blistering may occur in drug related pseudoporphyria, most commonly with NSAIDs. The severity of the reaction depends on the tissue level of the drug, its efficiency as a photosensitizer, and the extent of exposure to the activating wavelengths of ultraviolet (UV) light (Chap. 61). Common orally administered photosensitizing drugs include fluoroquinolones, tetracycline antibiotics, and trimethoprim/sulfamethoxazole. Other drugs less frequently implicated are chlorpromazine, thiazides, NSAIDs, and BRAF inhibitors. Voriconazole may result in severe photosensitivity, accelerated photoaging, and cutaneous carcinogenesis. Because UVA and visible light, which trigger these reactions, are not easily absorbed by nonopaque sunscreens and are transmitted through window glass, photosensitivity reactions may be difficult to block. Photosensitivity reactions abate with removal of either the drug or UV radiation, use of sunscreens that block UVA light, and treatment of the reaction as one would a sunburn. Rarely, individuals develop persistent reactivity to light, necessitating longterm avoidance of sun exposure. Some chemotherapeutic agents, such as methotrexate, can induce a Uvrecall reaction characterized by an erythematous, slightly scaly eruption at sites of prior severe sun exposure.

Dabrafenib–Trametinib Combination Approved for Solid Tumors with BRAF Mutations https://www.cancer.gov/news-events/cancer-currents-blog/2022/fda-dabrafenib-trametinib-braf-solid-tumors

Drug-Induced Photosensitivity-From Light and Chemistry to Biological Reactions and Clinical Symptoms

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Affiliations

- PMID: **34451820**
- PMCID: PMC8401619
- DOI: 10.3390/ph14080723

Drug-induced photosensitivity occurs when certain photosensitising medications cause unexpected sunburn or dermatitis (a dry, bumpy or blistering rash) on sun-exposed skin (face, neck, arms, backs of hands and often lower legs and feet). The rash may or may not be itchy. https://dermnetnz.org/topics/drug-induced-photosensitivity#:~:text=Drug-%20and%20chemical-induced%20photosensitivity%20occurs%20when%20a%20drug,topical%20agents%20or%20medications%20that%20are%20taken%20orally.

What are the clinical features of drug-induced photosensitivity?

The clinical features of drug-induced photosensitivity vary according to the photosensitising agent involved and the type of reaction it causes in the skin. The reaction can be phototoxic and/or photoallergic.

Phototoxic reactions result from direct damage to tissue caused by light activation of the photosensitising agent, whilst photoallergic reactions are a cell-mediated immune response in which the antigen is the light-activated photosensitising agent. Photoallergic reactions occur less commonly than phototoxic reactions and are mostly caused by photosensitising topical agents. Although some oral photosensitising medications can cause photoallergic reactions, most cause phototoxic reactions. A handful of medications can cause both phototoxic and photoallergic reactions.

The clinical features differ between phototoxic and photoallergic reactions.

Phototoxic reactions

- Skin reaction occurs minutes to hours after exposure to agent and light
- Appears as an exaggerated sunburn reaction (reddening and swelling)
- Vesicles, blisters and bullae may occur in severe reactions (pseudoporphyria)
- May or may not be itchy

•Less commonly, the skin may change colour, for example, a blue-green pigmentation is associated with amiodarone
•The reaction is limited to sun-exposed skin
•Photo-onycholysis (separation of the distal nail plate from the nail bed) may arise with many oral photosensitising medications and may be the only sign of phototoxicity in dark-skinned individuals

Photoallergic reactions

•Eczematous, itchy type reaction occurs 24-72 hours after exposure to agent and light
•May spread to areas that have not been sun-exposed
•Hyperpigmentation does not occur

Pigment Changes

- Causes:
  - Drugs that trigger melanocyte production of melanin
    - e.g., Oral contraceptives
  - Drugs that deposit metabolites
    - e.g., Amiodarone, minocycline
- Types and locations:
  - Various pigmentation patterns
    - bleomycin, busulfan, daunorubicin, cyclophosphamide, hydroxyurea, fluorouracil, and methotrexate.
  - Drug induced lipofuscinosis
    - Clofazimine (anti leprosy drug)
  - Hyperpigmentation of the face, mucous membranes, and pretibial and subungual areas
    - Pigmentation changes mucous membranes (busulfan, bismuth)
    - conjunctiva (chlorpromazine, thioridazine, imipramine, clomipramine),
    - nails (zidovudine, doxorubicin, cyclophosphamide, bleomycin, fluorouracil, hydroxyurea)
    - Hair (covered next)
    - Teeth (tetracyclines).

Pigmentation Changes

Drugs, either systemic or topical, may cause a variety of pigmentary changes in the skin by triggering melanocyte production of melanin (as in the case of oral contraceptives causing melasma) or due to deposition of drug or drug metabolites. Longterm minocycline and amiodarone may cause bluegray pigmentation. Phenothiazine, gold, and bismuth result in graybrown pigmentation of sunexposed areas. Numerous cancer chemotherapeutic agents may be associated with characteristic patterns of pigmentation (e.g., bleomycin, busulfan, daunorubicin, cyclophosphamide, hydroxyurea, fluorouracil, and methotrexate). Clofazimine causes a druginduced lipofuscinosis with characteristic redbrown coloration. Hyperpigmentation of the face, mucous membranes, and pretibial and subungual areas occurs with antimalarials. Quinacrine causes generalized yellow discoloration. Pigmentation changes may also occur in mucous membranes (busulfan, bismuth), conjunctiva (chlorpromazine, thioridazine, imipramine, clomipramine), nails (zidovudine, doxorubicin, cyclophosphamide, bleomycin, fluorouracil, hydroxyurea), hair, and teeth (tetracyclines).

Lipofuscin is the name **given to fine yellow-brown pigment granules composed of**

lipid-containing residues of lysosomal digestion. It is considered to be one of the aging or "wear-and-tear" pigments, found in the liver, kidney, heart muscle, retina, adrenals, nerve cells, and ganglion cells.

Hair

- Hair Loss
  - antineoplastic agents (alkylating agents, bleomycin, vinca alkaloids, platinum compounds), anticonvulsants (carbamazepine, valproate), beta blockers, antidepressants, antithyroid drugs, IFNs, oral contraceptives, and cholesterol lowering agents (e.g. atorvastatin 1% report).
- Hair Growth
  - Hirsutism: anabolic steroids, oral contraceptives, testosterone, corticotropin
  - Hypertrichosis –hair growth typically located on the forehead and temporal regions of the face. Anti-inflammatory drugs, glucocorticoids, vasodilators (diazoxide, minoxidil), diuretics (acetazolamide), anticonvulsants (phenytoin), immunosuppressive agents (cyclosporine A), psoralens, and zidovudine.
- Hair discoloration
  - chloroquine, IFNα, chemotherapeutic agents, and tyrosine kinase inhibitors. Changes in hair structure have been observed in patients given EGFR inhibitors, BRAF inhibitors, tyrosine kinase inhibitors, and acitretin.

Hypertrichosis

Chloroquine Hypopigmentation

Hirsutism

DrugInduced
Hair Disorders
DRUGINDUCED HAIR LOSS
Medications may affect hair follicles at two different phases of their growth cycle: anagen (growth) or telogen (resting). Anagen effluvium occurs within of drug administration, especially with antimetabolite or other chemotherapeutic drugs. In contrast, in telogen effluvium, the delay is 2–4 months following initiation of a new medication. Both present as diffuse, nonscarring alopecia most often reversible after discontinuation of the responsible agent. A considerable number of drugs have been associated with hair loss. These include antineoplastic agents (alkylating agents, bleomycin, vinca alkaloids, platinum compounds), anticonvulsants (carbamazepine, valproate), beta blockers, antidepressants, antithyroid drugs, IFNs, oral contraceptives, and cholesterollowering agents.

DRUGINDUCED HAIR GROWTH
Medications may also cause hair growth. Hirsutism is an excessive growth of terminal hair with masculine hair growth pattern in a female, most often on the face and trunk, due to androgenic stimulation of hormonesensitive hair follicles

(anabolic steroids, oral contraceptives, testosterone, corticotropin). Hypertrichosis is a distinct pattern of hair growth, not in a masculine pattern, typically located on the forehead and temporal regions of the face. Drugs responsible for hypertrichosis include anti-inflammatory drugs, glucocorticoids, vasodilators (diazoxide, minoxidil), diuretics (acetazolamide), anticonvulsants (phenytoin), immunosuppressive agents (cyclosporine A), psoralens, and zidovudine. Changes in hair color or structure are uncommon adverse effects from medications. Hair discoloration may occur with chloroquine, IFNα, chemotherapeutic agents, and tyrosine kinase inhibitors. Changes in hair structure have been observed in patients given EGFR inhibitors, BRAF inhibitors, tyrosine kinase inhibitors, and acitretin.

DrugInduced Nail Disorders

Drugrelated nail disorders usually involve all 20 nails and need months to resolve after withdrawal of the medication. The pathogenesis is most often toxic. Druginduced nail changes include Beau's line (transverse depression of the nail plate), onycholysis (detachment of the distal part of the nail plate), onychomadesis (detachment of the proximal part of the nail plate), pigmentation, and paronychia (inflammation of periungual skin).

ONYCHOLYSIS

Onycholysis occurs with tetracyclines, fluoroquinolones, retinoids, NSAIDs, and others, including many chemotherapeutic agents, and may be triggered by exposure to sunlight.

ONYCHOMADESIS

Onychomadesis is caused by temporary arrest of nail matrix mitotic activity. Common drugs reported to induce onychomadesis include carbamazepine, lithium, retinoids, and chemotherapeutic agents such as taxanes.

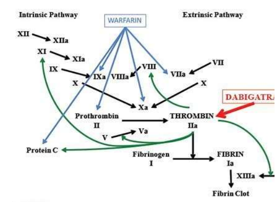
PARONYCHIA

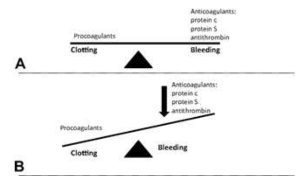
Paronychia and multiple pyogenic granulomas with progressive and painful periungual abscess of fingers and toes are side effects of systemic retinoids, lamivudine, indinavir, and antiEGFR monoclonal antibodies.

NAIL DISCOLORATION

Some drugs—including anthracyclines, taxanes, fluorouracil, psoralens, and zidovudine—may induce nail bed hyperpigmentation through melanocyte stimulation. It appears to be reversible and dose dependent.

Warfarin necrosis





- Warfarin-induced skin necrosis is a serious condition in which subcutaneous tissue necrosis occurs due to an acquired protein C deficiency following treatment with warfarin. The risk of necrosis increases in patients with protein C or protein S deficiency.

Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e

https://doi.org/10.1161/CIRCULATIONAHA.111.044412Circulation. 2011;124:e365–e368

This rare reaction (0.01–0.1%) usually occurs between the third and tenth days of therapy with warfarin, usually in women. Common sites are breasts, thighs, and buttocks (Fig. 601). Lesions are sharply demarcated, erythematous, or purpuric, and may progress to form large, hemorrhagic bullae with necrosis and eschar formation. areas adjacent to the injection site or more distant sites if infused. Levamisoletainted cocaine (and more recently, heroin) can induce similar skin necrosis; however, the distribution tends to involve the ears and cheeks predominantly, with stellate or retiform purpura. Patients may have abnormal white blood cell counts and may be dual Pand CANCA positive.

Adverse Effects

Serious adverse effects of warfarin include bleeding and significant hemorrhage. A significant hemorrhage (e.g., intracranial hemorrhage, gastrointestinal (GI) bleed, hematemesis, intraocular bleeding, hemarthrosis) can occur at virtually any site on the body. Patients should receive education about easy bleeding or bruising, which is a common adverse effect. A clinician should also counsel patients about the proper management of cuts, bruises, and nosebleeds. The risk of bleeding and hemorrhage is dependent on multiple variables, including the intensity of anticoagulation and

patient susceptibility. Individuals should undergo an assessment for their risk, with appropriate adjustments to their treatment plan made accordingly.
Other adverse effects include nausea, vomiting, abdominal pain, bloating, flatulence, and an altered sense of taste.
Rare cases of purple toe syndrome, warfarin-induced skin necrosis, and there are reports of calciphylaxis with warfarin therapy.[8][9] Purple toe syndrome is a complication characterized by cholesterol microembolization that causes purple lesions to develop on the toes and sides of the feet. Purple toe syndrome usually develops 3 to 8 weeks after the initiation of warfarin therapy. Warfarin-induced skin necrosis is a serious condition in which subcutaneous tissue necrosis occurs due to an acquired protein C deficiency following treatment with warfarin. The risk of necrosis increases in patients with protein C or protein S deficiency. The skin necrosis usually occurs within the first week of therapy and management strategies, including discontinuing treatment with warfarin, administering fresh frozen plasma and vitamin K, and initiating anticoagulation therapy with either unfractionated heparin or low molecular weight heparin. Calciphylaxis or calcium uremic arteriolopathy is another rare adverse effect in patients with or without end-stage renal disease. **Warfarin**
Shivali Patel; Ravneet Singh; Charles V. Preuss; Neepa Patel.
Author Information and Affiliations
Last Update: February 25, 2023.

https://my.clevelandclinic.org/health/diseases/21880-protein-c-deficiency

Circulation
Volume 124, Issue 14, 4 October 2011; Pages e365-e368
https://doi.org/10.1161/CIRCULATIONAHA.111.044412
Cardiology Patient Page
Deficiencies of Natural Anticoagulants, Protein C, Protein S, and Antithrombin

Toxic erythema of chemotherapy.

- Some cancer chemotherapy inhibit cell division --thus, rapidly proliferating elements of the skin, including hair, mucous membranes, and appendages, are sensitive to their effects.
  - Related skin effects include and are placed under the unifying diagnosis of toxic erythema of chemotherapy (TEC):
    - neutrophilic eccrine hidradenitis, sterile cellulitis, exfoliative dermatitis, and flexural erythema;
    - Acral erythema is marked by dysesthesia and an erythematous, edematous eruption of the palms and soles.
    - Common causes include cytarabine, doxorubicin, methotrexate, hydroxyurea, fluorouracil, and capecitabine.

Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e

Citation: Chapter 60 Cutaneous Drug Reactions, Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. *Harrison's Principles of Internal Medicine, 21e;* 2022. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=3095&sectionid=262791505 Accessed: April 10, 2023

Because many agents used in cancer chemotherapy inhibit cell division, rapidly proliferating elements of the skin, including hair, mucous membranes, and appendages, are sensitive to their effects. A broad spectrum of chemotherapy related skin toxicities has been reported, including neutrophilic eccrine hidradenitis, sterile cellulitis, exfoliative dermatitis, and flexural erythema; recent nomenclature classifies these under the unifying diagnosis of toxic erythema of chemotherapy (TEC) (Fig. 602). Acral erythema is marked by dysesthesia and an erythematous, edematous eruption of the palms and soles. Common causes include cytarabine, doxorubicin, methotrexate, hydroxyurea, fluorouracil, and capecitabine.

The recent introduction of many new monoclonal antibody and small molecular signaling inhibitors for the treatment of cancer has been accompanied by numerous reports of skin and hair toxicity; only the most common of these are mentioned here. EGFR antagonists induce follicular eruptions and nail toxicity after a mean interval of 10 days in a majority of patients. Xerosis, eczematous eruptions, acneiform eruptions, and pruritus are common. Erlotinib is associated with marked hair textural changes. Sorafenib, a tyrosine kinase inhibitor, may result in follicular eruptions and focal bullous eruptions at palmoplantar, flexural sites or areas of frictional pressure. BRAF inhibitors are associated with photosensitivity, palmoplantar hyperkeratosis, hair curling, dyskeratotic (Grover'slike) rash, hyperkeratotic benign cutaneous neoplasms, and keratoacanthomalike squamous cell carcinomas. Rash, pruritus, and vitiliginous depigmentation have been reported in association with ipilimumab (antiCTLA4) treatment. Up to 50% of patients experience immunemediated skin eruptions, including granulomatous reactions, dermatomyositis, panniculitis, and vasculitis. The checkpoint inhibitor class of drugs (including antiCTLA4, antiPD1, and antiPDL1 agents) can induce a wide range of cutaneous eruptions beyond vitiligo, including lichenoid, eczematous, granulomatous, papulosquamous, and panniculitis eruptions.

IMMUNE CUTANEOUS REACTIONS: COMMON
Maculopapular Eruptions

- Common offenders:
  - aminopenicillins, cephalosporins, antibacterial sulfonamides, allopurinol, and antiepileptic drugs. Beta blockers, calcium channel blockers, and ACE inhibitors may occur.
  - High risk drugs include: nevirapine and lamotrigine (associated with higher starting doses and use in children)
- Maculopapular reactions usually develop within 1 week of initiation of therapy and last less than 2 weeks.
- Suspect drug should be discontinued unless it is essential.
  - Note: Rash may continue to progress for a few days up to 1 week following medication discontinuation.
- Oral antihistamines and emollients may help relieve pruritus. Short courses of potent topical glucocorticoids can reduce inflammation and symptoms. Systemic glucocorticoid treatment is rarely indicated.

Morbilliform drug eruption.

Citation: Chapter 60 Cutaneous Drug Reactions, Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. *Harrison's Principles of Internal Medicine, 21e;* 2022. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=3095&sectionid=262791505 Accessed: April 10, 2023

Morbilliform or maculopapular eruptions (Fig. 603) are the most common of all drugninduced reactions, often start on the trunk or intertriginous areas, and consist of blanching erythematous macules and papules that are symmetric and confluent. Nonblanching, dusky, or bright red macules as well as mucosal involvement should raise concern for a more severe reaction. Facial involvement in morbilliform eruptions is also uncommon, and the presence of extensive facial lesions with facial edema suggests DIHS. Diagnosis of morbilliform eruptions is rarely assisted by laboratory testing or skin biopsy.

Morbilliform eruptions may be associated with moderate to severe pruritus and fever. A viral exanthem is another differential diagnostic consideration, especially in children, and graftversushost disease is also a consideration in the proper clinical setting. Absence of enanthems; absence of ear, nose, throat, and upper respiratory tract symptoms; and polymorphism of the skin lesions support a drug rather than a viral eruption. Common offenders include aminopenicillins, cephalosporins, antibacterial sulfonamides, allopurinol, and antiepileptic drugs. Beta blockers, calcium channel blockers, and ACE inhibitors are rarely the culprit; however, any drug can cause a morbilliform exanthem. Certain medications carry very high rates of morbilliform eruption, including nevirapine and lamotrigine, even in the absence of DIHS reactions. Lamotrigine morbilliform rash is associated with higher starting doses, rapid dose escalation, concomitant use of valproate (which increases lamotrigine levels and halflife), and use in children. Maculopapular reactions usually develop within 1 week of initiation of therapy and last less than 2 weeks. Occasionally, these eruptions resolve despite continued use of the responsible drug. Because the eruption may also worsen, the suspect drug should be discontinued unless it is essential. It is important to note that the rash may continue to progress for a few days up to 1 week following medication discontinuation. Oral antihistamines and emollients may help relieve pruritus. Short courses of potent topical glucocorticoids can reduce inflammation and symptoms. Systemic glucocorticoid treatment is rarely indicated.

Pruritus and Urticaria

- Pruritus
  - Most common drug eruptions.
  - Antihistamines such as hydroxyzine or diphenhydramine help.
  - Pruritus stemming from specific medications may require distinct treatment, such as selective opiate antagonists for opiate related pruritus.
- Urticaria/Angioedema/Anaphylaxis
  - Urticaria, the second most frequent
  - characterized by pruritic, red wheals of varying size rarely lasting more than 24 hours.
    - association with nearly all drugs, most frequently ACE inhibitors, aspirin, NSAIDs, penicillin, and blood products.
    - Angiodema = Deep edema within dermal and subcutaneous tissues may also involve respiratory and gastrointestinal mucous membranes. Urticaria and angioedema may be part of a life-threatening anaphylactic reaction.
- Mechanism include
  - IgE dependent (IgE dependent urticarial reactions -- ~ 36 hours of drug exposure)
  - circulating immune complexes (serum sickness) (reactions usually occurs 6–12 days after first exposure)
  - nonimmunologic activation of effector pathways.

- Anaphylactoid Reactions
  - Vancomycin --red man syndrome, a histamine-related anaphylactoid reaction characterized by flushing, diffuse maculopapular eruption, and hypotension. In rare cases, cardiac arrest may be associated with rapid intravenous (IV) infusion of the medication.

Pruritus

Pruritus is associated with almost all drug eruptions and, in some cases, may represent the only symptom of the adverse cutaneous reaction. It may be alleviated by antihistamines such as hydroxyzine or diphenhydramine. Pruritus stemming from specific medications may require distinct treatment, such as selective opiate antagonists for opiate related pruritus.

Urticaria/Angioedema/Anaphylaxis

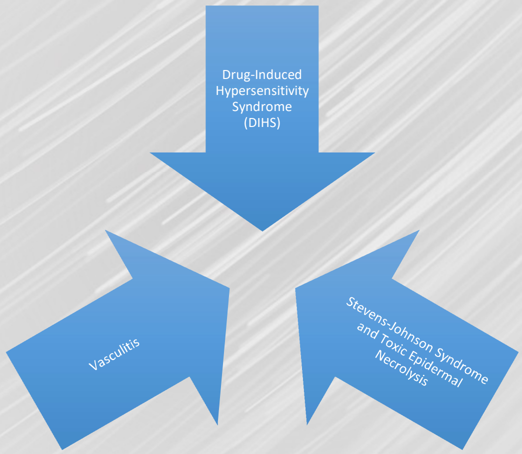
Urticaria, the second most frequent type of cutaneous reaction to drugs, is characterized by pruritic, red wheals of varying size rarely lasting more than 24 hours. It has been observed in association with nearly all drugs, most frequently ACE inhibitors, aspirin, NSAIDs, penicillin, and blood products. However, medications account for no more than 10–20% of acute urticaria cases. Deep edema within dermal and subcutaneous tissues is known as angioedema and may involve respiratory and gastrointestinal mucous membranes. Urticaria and angioedema may be part of a lifethreatening anaphylactic reaction.

Drug induced urticaria may be caused by three mechanisms: an IgE dependent mechanism, circulating immune complexes (serum sickness), and nonimmunologic activation of effector pathways. IgE dependent urticarial reactions usually occur within 36 hours of drug exposure but can occur. within minutes. Immune complex–induced urticaria associated with serum sickness reactions usually occurs 6–12 days after first exposure. In this syndrome, the urticarial eruption (typically polycyclic plaques over distal joints) may be accompanied by fever, hematuria, arthralgias, hepatic dysfunction, and neurologic symptoms. Certain drugs, such as NSAIDs, ACE inhibitors, angiotensin II antagonists, radiographic dye, and opiates, may induce urticarial reactions, angioedema, and anaphylaxis in the absence of drug specific antibodies through direct mast-cell degranulation. Radiocontrast agents are a common cause of urticaria and, in rare cases, can cause anaphylaxis. High-osmolality radiocontrast media are about five times more likely to induce urticaria (1%) or anaphylaxis than are newer low-osmolality media. About one-third of those with mild reactions to previous exposure react on re-exposure. Pretreatment with prednisone and diphenhydramine reduces reaction rates. The treatment of urticaria or angioedema depends on the severity of the reaction. In severe cases with respiratory or cardiovascular compromise, epinephrine and intravenous glucocorticoids are the mainstay of therapy. For patients with urticaria without symptoms of angioedema or anaphylaxis, drug withdrawal and oral antihistamines are usually sufficient. Future drug avoidance is recommended; rechallenge, especially in individuals with severe reactions, should only occur in an intensive care setting.

Anaphylactoid Reactions

Vancomycin is associated with red man syndrome, a histaminerelated anaphylactoid reaction characterized by flushing, diffuse maculopapular eruption, and hypotension. In rare cases, cardiac arrest may be associated with rapid intravenous (IV) infusion of the medication.

IMMUNE CUTANEOUS REACTIONS: RARE AND SEVERE



Drug-induced hypersensitivity syndrome/drug rash with eosinophilia and systemic symptoms (DIHS/DRESS).

The risk of DIHS has been estimated between 1 in 1000 to 1 in 10,000 drug exposures, though the true incidence of the syndrome is unknown.

Pathophysiology – drug metabolic enzymes and HLA haplotypes.

Symptoms: prodrome of fever and flulike symptoms for several days, followed by the appearance of a diffuse morbilliform eruption, usually involving the face. Facial swelling and hand/foot swelling are often present. Systemic manifestations include lymphadenopathy, fever, and leukocytosis (often with eosinophilia or atypical lymphocytosis), as well as hepatitis, nephritis, pneumonitis, myositis, and gastroenteritis, in descending order.

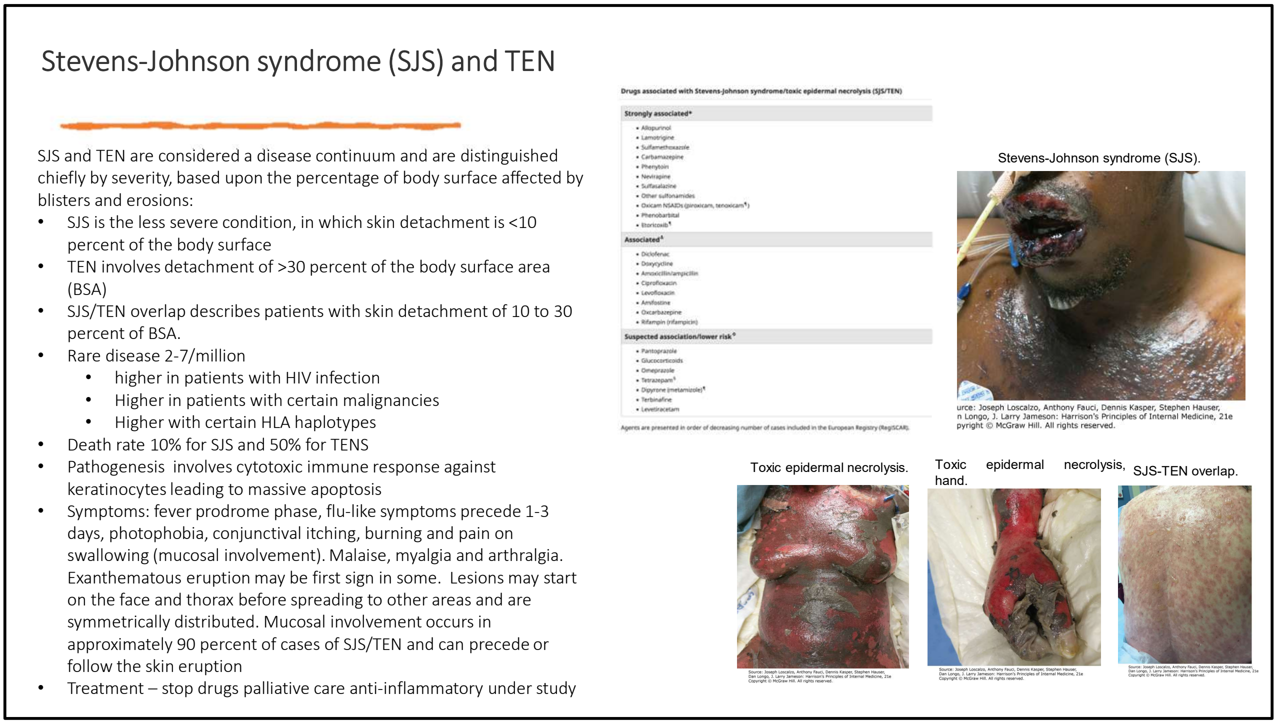
- Onset of timing and organ affects depends on drug.
  - **Allopurinol** classically induces DIHS with renal involvement;
  - **Minocycline involve heart and lungs**
  - **Abacavir** involve GI
  - DIHS is a preferred term -- some medications typically do not induce eosinophilia **(abacavir, dapsone, lamotrigine).**
  - Crossreactions are common e.g. **aromatic anticonvulsants, including phenytoin, carbamazepine, and phenobarbital**
  - Other drugs causing DIHS include **antibacterial sulfonamides and other antibiotics**.
  - Drug metabolites may be involved.
- The cutaneous reaction usually begins 2–8 weeks after the drug is started and persists after drug cessation. Signs and symptoms may continue for several weeks, especially those associated with hepatitis. The eruption recurs with rechallenge, and crossreactions among, are common
- Mortality rates as high as 10% have been reported, with most fatalities resulting from liver failure.
- Treatment: systemic glucocorticoids (1.5–2 mg/kg/d prednisone equivalent) should be started and tapered slowly over 8–12 weeks, CBC, LP, basic metabolic panel are important. A steroid-sparing agent such as mycophenolate mofetil, IV immunoglobulin, or cyclosporine may be indicated in cases of rapid recurrence upon steroid taper.
- In all cases, immediate withdrawal of the suspected culprit drug is required.
- Organ function must be monitored for long term affects.

https://utahderm.med.utah.edu/diagnoses/drug-induced-hypersensitivity-syndrome-dihs-and-drug-reaction-with-eosinophilia-and-systemic-symptoms-dress-syndrome/

Citation: Chapter 60 Cutaneous Drug Reactions, Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. *Harrison's Principles of Internal Medicine, 21e;* 2022. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=3095&sectionid=262791505 Accessed: April 10, 2023

Drug-Induced Hypersensitivity Syndrome

DIHS is a systemic drug reaction also known as DRESS (drug reaction with eosinophilia and systemic symptoms) syndrome; because eosinophilia is not always present, the term DIHS is preferred. Clinically, DIHS presents with a prodrome of fever and flulike symptoms for several days, followed by the appearance of a diffuse morbilliform eruption, usually involving the face (Fig. 606). Facial swelling and hand/foot swelling are often present. Systemic manifestations include lymphadenopathy, fever, and leukocytosis (often with eosinophilia or atypical lymphocytosis), as well as hepatitis, nephritis, pneumonitis, myositis, and gastroenteritis, in descending order. Distinct patterns of timing of onset and organ involvement may exist. For example, allopurinol classically induces DIHS with renal involvement; cardiac and lung involvement are more common with minocycline; gastrointestinal involvement is almost exclusively seen with abacavir; and some medications typically do not induce eosinophilia (abacavir, dapsone, lamotrigine). The cutaneous reaction usually begins 2–8 weeks after the drug is started and persists after drug cessation. Signs and symptoms may continue for several weeks, especially those associated with hepatitis. The eruption recurs with rechallenge, and crossreactions among aromatic anticonvulsants, including phenytoin, carbamazepine, and phenobarbital, are common. Other drugs causing DIHS include antibacterial sulfonamides and other antibiotics. Hypersensitivity to reactive drug metabolites, hydroxylamine for sulfamethoxazole and arene oxide for aromatic anticonvulsants, may be involved in the pathogenesis of DIHS. Recent research suggests that inciting drugs may reactivate quiescent human herpes viruses, including herpesviruses 6 and 7, EBV, and cytomegalovirus (CMV), resulting in expansion of viralspecific CD8+ T lymphocytes and subsequent endorgan damage. Viral reactivation may be associated with a worse clinical prognosis. Mortality rates as high as 10% have been reported, with most fatalities resulting from liver failure. Systemic glucocorticoids (1.5–2 mg/kg/d prednisone equivalent) should be started and tapered slowly over 8–12 weeks, during which time clinical symptoms and labs (including complete blood count with differential, basic metabolic panel, and liver function tests) should be followed carefully. A steroid-sparing agent such as mycophenolate mofetil, IV immunoglobulin, or cyclosporine may be indicated in cases of rapid recurrence upon steroid taper. In all cases, immediate withdrawal of the suspected culprit drug is required. Given the severe long-term complications of myocarditis, patients should undergo cardiac evaluation in cases of severe DIHS or if heart involvement is suspected due to hypotension or arrhythmia. Patients should be closely monitored for resolution of organ dysfunction and for development of late-onset autoimmune



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SJS and TEN are characterized by blisters and mucosal/epidermal detachment resulting from fullthickness epidermal necrosis in the absence of substantial dermal inflammation. The term StevensJohnson syndrome (SJS) describes cases in which the total body surface area of blistering and eventual detachment is <10% (Fig. 607).
The term StevensJohnson syndrome/toxic epidermal necrolysis (SJS/TEN) overlap is used to describe cases with 10–30% epidermal detachment (Fig. 608), and the term toxic epidermal necrolysis (TEN) is used to describe cases with >30% detachment (Figs. 609 and 6010).

Other blistering eruptions with concomitant mucositis may be confused with SJS/TEN. Erythema multiforme (EM) associated with herpes simplex virus is characterized by painful mucosal erosions and target lesions, typically with an acral distribution and limited skin detachment. Mycoplasma and other respiratory infections in children cause a clinically distinct presentation with prominent mucositis and limited cutaneous involvement. The term reactive infectious mucocutaneous eruption (RIME) has been proposed to help differentiate this clinical entity, which some believe may be the syndrome originally described by Stevens and Johnson. Patients with SJS/TEN initially present with fever >39°C (102.2°F); sore throat; conjunctivitis; and acute onset of painful dusky, atypical, targetlike lesions (Fig. 6011). Intestinal and upper respiratory tract involvement are associated with a poor prognosis, as are older age and greater extent of epidermal detachment. At least 10% of those with SJS and 30% of those with TEN die from the disease. Drugs that most commonly cause SJS/TEN are sulfonamides, allopurinol, antiepileptics (e.g., lamotrigine, phenytoin, carbamazepine), oxicam NSAIDs, βlactam and other antibiotics, and nevirapine. Frozensection skin biopsy may aid in rapid diagnosis.

At this time, there is no consensus on the most effective treatment for SJS/TEN. The best outcomes stem from early diagnosis, immediate discontinuation of the suspected drug, and meticulous supportive therapy in an intensive care or burn unit. Fluid management, atraumatic wound care, infection prevention and treatment, and ophthalmologic and respiratory support are critical. Early administration of systemic glucocorticoids, intravenous immunoglobulin, cyclosporine, or etanercept may improve disease outcomes, but randomized studies to evaluate potential therapies are lacking and difficult to perform.

Who is at risk of developing Stevens-Johnson syndrome?

Some groups of people are at a higher risk of getting Stevens-Johnson syndrome, including:

- People who have experienced certain symptoms after taking specific medications
- Those with a weakened immune system
- Individuals with AIDS or HIV
- People undergoing chemotherapy
- Those who have family members with Stevens-Johnson syndrome

Stevens-Johnson Syndrome Causes

The most common cause of SJS is an adverse drug reaction. Almost any drug can result in SJS, but medicines such as antibiotics, anticonvulsants and anti-inflammatory treatments most frequently cause it. SJS is more common in children and younger adults, but can develop at any age. Typical eye problems associated with SJS can include:

- Conjunctivitis

•Ulceration of the eyelids
•Inflammation inside the eye (iritis)
•Corneal blisters
•Corneal perforation, a hole in the cornea that can lead to permanent vision loss
After the acute stage of the disease, conjunctival and corneal scarring, which lead to decrease or loss of vision, are common.
In children, Stevens-Johnson syndrome can also be caused by infections such as cold or flu, or cold sores.
Stevens-Johnson Syndrome Symptoms
Early stage Steven-Johnson syndrome can include flu-like symptoms, such as fever, sore throat, cough or joint pain. A few days later, a skin rash may appear on any part of the body and may feel like sunburn or lesions. Other symptoms, which appear shortly after the initial issues, may include:
•Blisters on the skin, mouth, eyes, nose and genitals
•Shedding skin after the blisters form
•Unexplained pain on the skin
•Swelling of the mouth, lips, throat, tongue or face
Stevens-Johnson Syndrome Diagnosis
Diagnosis of SJS is made by an ophthalmologist or dermatologist during a thorough medical evaluation that includes a physical exam and review of medication history. A skin biopsy may be performed by a dermatologist for exact diagnosis.
An important element of the early diagnosis is high degree of suspicion by emergency room providers after gathering medication history. The providers will then ask for ophthalmology and dermatology consults. It is important to immediately stop use of medications while the care team determines the cause of the inflammation.
Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis
Occasionally, Stevens-Johnson syndrome can develop into toxic epidermal necrolysis. SJS is when the inflammation affects less than one-tenth of the body surface area. When the lesions cover about one-third or more of the body surface area, it is considered toxic epidermal necrolysis.
Stevens-Johnson Syndrome Treatment
Stevens-Johnson syndrome needs to be treated in the hospital, usually in an intensive care unit. These treatments can include:
•IV fluids to prevent dehydration from skin loss.
•Treatment of affected areas such as blisters on the skin. Cold compress, creams and bandages can help protect the skin, and medications can control inflammation, manage pain and prevent additional infection. Special wound dressing or amniotic membranes may be required for significant cases.
•Intravenous immunoglobulin treatment, or high dose steroids might be helpful at early stages to limit the disease.
•Care for eye inflammation starts by using artificial tears and ointments to lubricate the ocular surface. Steroid drops should be used in patients with significant

conjunctival redness and inflammation. Antibiotic drops and ointment are added to regimens in patients with ulcerations, which can be diagnosed through eye examinations. Corneal transplants, limbal stem cell transplantations or artificial corneal procedures may be considered in the later stages of the disease, after the acute inflammation, if advised by an ophthalmologist.

Stevens-Johnson Syndrome Complications

Seeking immediate medical attention for Stevens-Johnson syndrome is important. The reaction can continue during the use of the causative medication and up to two weeks after stopping. In addition, some people may experience complications such as:

•Dehydration: Sores in the throat and mouth may make fluid intake challenging, and areas where the skin has shed tend to lose fluids through evaporation.

•Sepsis: Blood infection or sepsis occurs when bacteria enters the bloodstream from the open sores in the skin and spreads throughout the body. If left untreated, it can cause shock and/or organ failure.

•Eye problems: If the rash from SJS spreads to the eye, it can lead to inflammation, light sensitivity, redness or painful ulcers in the conjunctiva or cornea. In chronic stages, dry eyes and vision impairment due to corneal involvement can occur.

•Respiratory distress: In some cases, Stevens-Johnson syndrome can lead to a situation in which the lungs aren't able to transport enough oxygen into the blood, leading to wheezing, tightness in the throat or chest, and not being able to breathe or talk.

Frequently, after the skin grows back, scars, bumps or unusual coloring may remain. Some patients may experience loss of hair, or toenails and fingernails not growing as they did before.

Cutaneous small-vessel vasculitis (CSVV, leukocytoclastic vasculitis).

Cutaneous small-vessel vasculitis (CSVV)

Cutaneous smallvessel vasculitis (CSVV) typically presents with purpuric papules and macules involving the lower extremities and other dependent areas (Fig. 6013) (Chap. 363). Pustular and hemorrhagic vesicles as well as rounded ulcers also occur. Importantly, vasculitis may involve other organs, including the kidneys, joints, gastrointestinal tract, and lungs, necessitating a thorough clinical evaluation for systemic involvement. Drugs are implicated as a cause of roughly 15% of all cases of small vessel vasculitis. Antibiotics, particularly βlactams, are commonly implicated; however, almost any drug can cause vasculitis. Vasculitis may also be idiopathic or due to underlying infection, connective tissue disease, or (rarely) malignancy.

Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e

Citation: Chapter 60 Cutaneous Drug Reactions, Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. *Harrison's Principles of Internal Medicine, 21e;* 2022. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=3095&sectionid=262791505 Accessed: April 10, 2023

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Management

- Early Diagnosis
  - Clinical Laboratory findings
  - Clinical Features of Severe Cutaneous Drug Reactions

There are four main questions to answer regarding a suspected drug eruption:
1. Is the observed rash caused by a medication?
2. Is the reaction severe or evolving with systemic involvement?
3. Which drug or drugs are suspected, and should they be withdrawn?
4. What recommendation can be made for future medication use?

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Clinical Findings

TABLE 60-2

Clinical and Laboratory Findings Suggestive of Severe Cutaneous Adverse Drug Reaction

Cutaneous
Generalized erythema Facial edema Skin pain Palpable purpura Dusky or target-like lesions Skin necrosis Blisters or epidermal detachment Positive Nikolsky sign Mucous membrane erosions Swelling of lips or tongue
General
High fever Enlarged lymph nodes Arthralgias or arthritis Shortness of breath, hoarseness, wheezing, hypotension
Laboratory Results
Eosinophil count >1000/μL Lymphocytosis with atypical lymphocytes Abnormal liver or kidney function tests

Source: From JC Roujeau, RS Stern: Severe adverse cutaneous reactions to drugs. N Engl J Med 331:1272, 1994.

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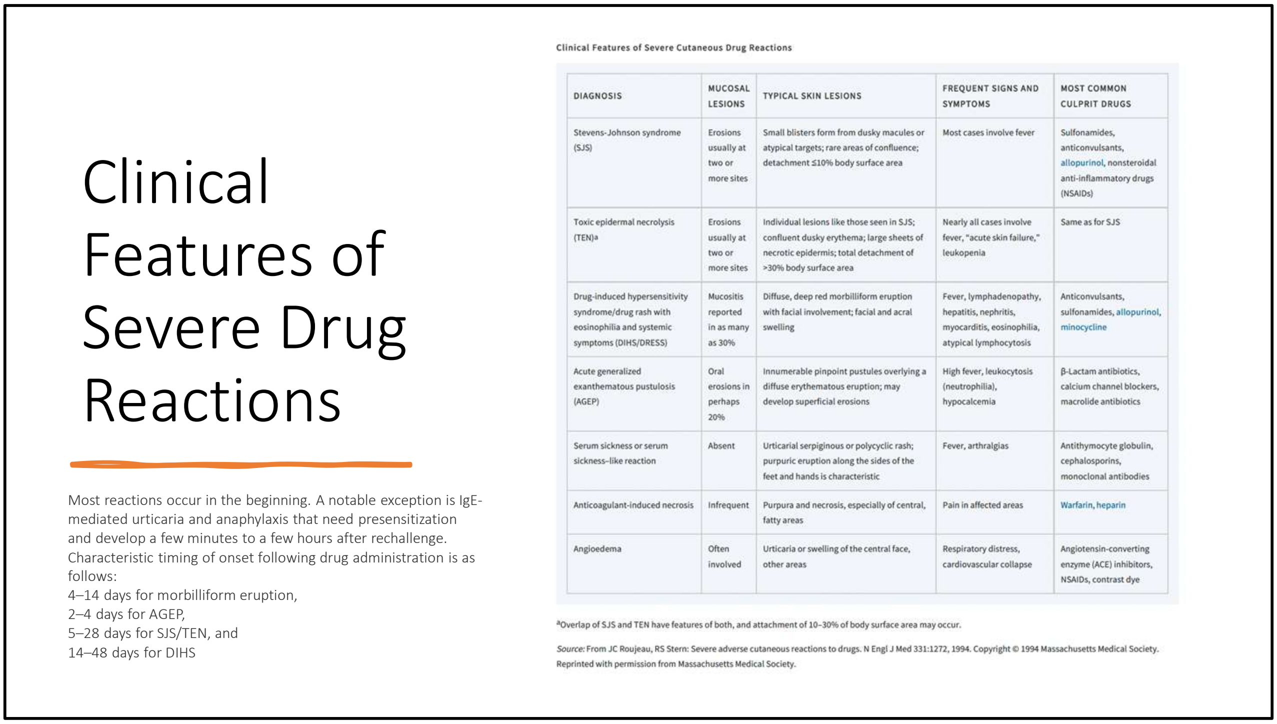

Clinical Features of Severe Drug Reactions

Most reactions occur in the beginning. A notable exception is IgE-mediated urticaria and anaphylaxis that need presensitization and develop a few minutes to a few hours after rechallenge. Characteristic timing of onset following drug administration is as follows:
4–14 days for morbilliform eruption,
2–4 days for AGEP,
5–28 days for SJS/TEN, and
14–48 days for DIHS

Clinical Features of Severe Cutaneous Drug Reactions

DIAGNOSIS	MUCOSAL LESIONS	TYPICAL SKIN LESIONS	FREQUENT SIGNS AND SYMPTOMS	MOST COMMON CULPRIT DRUGS
Stevens-Johnson syndrome (SJS)	Erosions usually at two or more sites	Small blisters form from dusky macules or atypical targets; rare areas of confluence; detachment ≤10% body surface area	Most cases involve fever	Sulfonamides, anticonvulsants, allopurinol, nonsteroidal anti-inflammatory drugs (NSAIDs)
Toxic epidermal necrolysis (TEN)[a]	Erosions usually at two or more sites	Individual lesions like those seen in SJS; confluent dusky erythema; large sheets of necrotic epidermis; total detachment of >30% body surface area	Nearly all cases involve fever, "acute skin failure," leukopenia	Same as for SJS
Drug-induced hypersensitivity syndrome/drug rash with eosinophilia and systemic symptoms (DIHS/DRESS)	Mucositis reported in as many as 30%	Diffuse, deep red morbilliform eruption with facial involvement; facial and acral swelling	Fever, lymphadenopathy, hepatitis, nephritis, myocarditis, eosinophilia, atypical lymphocytosis	Anticonvulsants, sulfonamides, allopurinol, minocycline
Acute generalized exanthematous pustulosis (AGEP)	Oral erosions in perhaps 20%	Innumerable pinpoint pustules overlying a diffuse erythematous eruption; may develop superficial erosions	High fever, leukocytosis (neutrophilia), hypocalcemia	β-Lactam antibiotics, calcium channel blockers, macrolide antibiotics
Serum sickness or serum sickness–like reaction	Absent	Urticarial serpiginous or polycyclic rash; purpuric eruption along the sides of the feet and hands is characteristic	Fever, arthralgias	Antithymocyte globulin, cephalosporins, monoclonal antibodies
Anticoagulant-induced necrosis	Infrequent	Purpura and necrosis, especially of central, fatty areas	Pain in affected areas	Warfarin, heparin
Angioedema	Often involved	Urticaria or swelling of the central face, other areas	Respiratory distress, cardiovascular collapse	Angiotensin-converting enzyme (ACE) inhibitors, NSAIDs, contrast dye

[a]Overlap of SJS and TEN have features of both, and attachment of 10–30% of body surface area may occur.

Source: From JC Roujeau, RS Stern: Severe adverse cutaneous reactions to drugs. N Engl J Med 331:1272, 1994. Copyright © 1994 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

CONFIRMATION OF DRUG REACTION

The probability of drug etiology varies with the pattern of the reaction. Only fixed drug eruptions are always drug-induced. Morbilliform eruptions are usually viral in children and drug-induced in adults. Among severe reactions, drugs account for 10–20% of anaphylaxis and vasculitis and between 70% and 90% of AGEP, DIHS, SJS, and TEN. Skin biopsy helps characterize the reaction but does not indicate drug causality. Blood counts and liver and renal function tests are important for evaluating organ involvement. The association of mild elevation of liver enzymes and high eosinophil count is frequent but not specific for a drug reaction. Blood tests that could identify an alternative cause, serologic tests (to rule out drug-induced lupus), and serology or polymerase chain reaction for infections may be of great importance to determine a cause.

WHAT DRUG(S) TO SUSPECT AND WITHDRAW

Most cases of drug eruptions occur during the first course of treatment with a new medication. A notable exception is IgE-mediated urticaria and anaphylaxis that need presensitization and develop a few minutes to a few hours after rechallenge. Characteristic timing of onset following drug administration is as follows: 4–14 days for morbilliform eruption, 2–4 days for AGEP, 5–28 days for

SJS/TEN, and 14–48 days for DIHS. A drug chart, compiling information of all current and past medications/supplements and the timing of administration relative to the rash, is a key diagnostic tool for identifying the inciting drug. Medications introduced for the first time in the relevant time frame are prime suspects. Two other important elements to suspect causality at this stage are (1) previous experience with the drug (or related members of the same pharmacologic class) and (2) alternative etiologic candidates. The decision to continue or discontinue any medication depends on the severity of the reaction, the severity of the primary disease undergoing treatment, the degree of suspicion of causality, and the feasibility of finding an alternative safer treatment. In any potentially fatal drug reaction, elimination of all possible suspect drugs or unnecessary medications should be immediately attempted. Some rashes may resolve when “treating through” a benign drugrelated eruption. The decision to treat through an eruption should, however, remain the exception and withdrawal of every suspect drug the general rule. On the other hand, drugs that are not suspected and are important for the patient (e.g., antihypertensive agents) generally should not be quickly withdrawn. This approach may permit judicious use of these agents in the future.

RECOMMENDATION FOR FUTURE USE OF DRUGS

The aims are to (1) prevent the recurrence of the drug eruption and (2) avoid compromising future treatment by inaccurately excluding otherwise useful medications. A thorough assessment of drug causality is based on timing of the reaction, evaluation of other possible causes, and effect of drug withdrawal or continuation. The RegiSCAR group has proposed the Algorithm of Drug Causality for Epidermal Necrolysis (ALDEN) to rank likelihood of drug causality in SJS/TEN; validation of this and other instruments, such as the Naranjo adverse drug reaction probability scale, is limited. Medication(s) with a “definite” or “probable” causality should be contraindicated, a warning

Good Luck

- Objectives:
  - Describe the non-immunologic types of cutaneous drug reactions
  - Describe the immunologic types of cutaneous drug reactions
  - Be able to explain to your patients the advantages of genetic screening related to cutaneous adverse effects.
  - Recognize the cutaneous presentations of adverse drug effects and relate them to specific drugs.
  - Describe warfarin necrosis to your patient.
  - Know the drugs that can exacerbate existing cutaneous disease.
  - Explain to your patient what photosensitivity is and what might cause this reaction.
  - Be able to talk to your patients about drugs that cause hair loss or growth.
  - Explain to your patient the pigmentation changes that may occur with certain medications.
- Recognize the clinical manifestations of common and severe cutaneous drug reactions including rug-Induced Hypersensitivity Syndrome (DIHS)
  - Common rashes
  - Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis
  - Vasculitis
- Know the most common drugs associated with both common and severe cutaneous reactions.
  - **Immediate Immune Reactions:** **NSAIDs (know common ones), including aspirin, and radiocontrast media**. **Penicillin's (know common ones)** and **muscle relaxants** used in general anesthesia (**succinylcholine, vecuronium, pancuronium and atracurium)**
  - **Immune complex dependent reactions:** **Penicillin, cefaclor, amoxicillin, trimethoprim/sulfamethoxazole**, and monoclonal antibodies such as **infliximab, rituximab**, and **omalizumab**
  - **Delayed hypersensitivity reactions:** Sulfonamides, anticonvulsants, allopurinol, nonsteroidal, anti-inflammatory drugs (NSAIDs), Anticonvulsants, sulfonamides, allopurinol, minocycline, βLactam antibiotics, calcium channel blockers, macrolide antibiotics, Antithymocyte globulin, cephalosporins, monoclonal antibodies, Angiotensinconverting enzyme (ACE) inhibitors, NSAIDs, contrast dye
  - **Drugs that produce anti-coagulant induced necrosis:** Warfarin, heparin
  - **Drugs that may exacerbate existing cutaneous diseases: NSAIDs, lithium, beta blockers, tumor necrosis factor (TNF) antagonists, interferon (IFN) α, and ACE inhibitors** Antimalarials or glucocorticoid withdrawal, glucocorticoids, androgens, lithium, and antidepressants, thiazide diuretics, proton pump inhibitors, TNF inhibitors, terbinafine, and minocycline, TNF inhibitors or capecitabine; hydroxyurea
  - **Drugs that are associated with photo-sensivity reactions:** fluoroquinolones, tetracycline antibiotics, and trimethoprim/sulfamethoxazole. Other drugs less frequently implicated are chlorpromazine, thiazides, NSAIDs, and BRAF inhibitors, voriconazole
  - **Drugs that may cause pigmentation changes: Oral contraceptives, Amiodarone, minocycline, bleomycin, busulfan, daunorubicin, cyclophosphamide, hydroxyurea, fluorouracil, and methotrexate, busulfan, bismuth, chlorpromazine, thioridazine, imipramine, clomipramine, zidovudine, doxorubicin, cyclophosphamide, bleomycin, fluorouracil, hydroxyurea, tetracyclines**
  - Drugs that affect hair growth and color: antineoplastic agents (**alkylating agents, bleomycin, vinca alkaloids, platinum compounds)**, anticonvulsants (**carbamazepine, valproate**), **beta blockers, antidepressants, antithyroid drugs, IFNs, oral contraceptives, and cholesterol lowering agents (e.g. atorvastatin 1% report), anabolic steroids, oral contraceptives, testosterone, corticotropin, Anti-inflammatory drugs, glucocorticoids**, vasodilators (**diazoxide, minoxidil**), diuretics (**acetazolamide**), anticonvulsants (phenytoin), immunosuppressive agents (**cyclosporine A**), psoralens, and zidovudine. **Chloroquine, IFNα, chemotherapeutic agents, and tyrosine kinase inhibitors**. EGFR inhibitors, BRAF inhibitors, tyrosine kinase inhibitors, and acitretin.